

CELLCHECK · REQUIREMENTS

CellCheck requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CCK-REQ-001	Version	0.6
Status	Draft	Date	2026-10-01
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cellcheck		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; cost wording only, no number changed
0.5	2026-10-01	Amish Chadha	R11 and R12 values for the constructable design (CCK-DDR-003, CCK-CAL-001 v0.3); no status changed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R12 target \$175 and met; R4 met with the temperature-gated pulse; R6 read in the same channel; status table from CCK-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Apply CCK-DDR-001 (R14 redefined, R9 protection adopted, R3 calibration, R12 against both budgets); status table from CCK-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

CellCheck requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be revised with the first trial partner. Status in Table 2 comes from the calculation note CCK-CAL-001 v0.3. No requirement is failed outright; three are at risk (R5, R6 and R11), and R10 cannot be verified at TRL 3. R14 was redefined in v0.3 under CCK-DDR-001 item 2, and R12 was restated in v0.4 for the \$175 value-engineering target; both follow decisions Amish accepted on 2026-09-25 (CCK-DDR-002).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Test the common salvaged cell sizes	18650 and 21700 cylindrical lithium-ion cells (LCO, NMC, NCA), one cell per channel, 8 channels or more	Design review of holders and profiles
R2	Measure capacity accurately	± 2 % of reading at a constant 1.0 A discharge from 4.20 V to 2.80 V	Error budget; later comparison with a calibrated reference
R3	Measure cell voltage accurately	± 10 mV from 2.0 to 4.3 V, after a one-time per-channel calibration at build	Data sheet error budget; later comparison with a calibrated meter
R4	Measure DC internal resistance repeatably	Resolution 1 m Ω or better; repeatability ± 3 m Ω or ± 5 % of reading, whichever is larger, over 5 reinsertions of one cell	Error budget; later repeat test
R5	Grade enough cells	12 cells or more per attended 10 h day on typical 1.8 Ah cells	Cycle-time calculation
R6	Detect self-discharge	Open-circuit voltage change over a 14-day rest measured to ± 3 mV, with both readings in the same channel; cells dropping more than 50 mV (default) beyond the median of the same-day cohort flagged	Calculation; later repeat readings
R7	Monitor cell temperature	Each cell measured at least once per second to ± 2 K; channel stops above 45 °C or at a rise of more than 8 K above the bench	Design review; later bench test
R8	Keep the unit cool at full load	Load MOSFET cases at 75 °C or less with all channels discharging at 1 A in a 35 °C room	Thermal calculation
R9	Stay safe after a single fault	No single failure (controller crash, stuck-on load MOSFET, open thermistor, failed charger) lets a cell be charged above 4.25 V, heated past 60 °C, or discharged below 2.5 V	Failure mode and effects analysis
R10	Contain a single cell failure	Tray, liner and guard hold ejecta and flame from one venting 21700 cell without spread beyond the tray	Analysis at TRL 3; physical test only at TRL 4 or later
R11	Fit on a bench	500 x 320 x 120 mm or smaller, 5 kg or less without the adapter	Massing model; later weighing
R12	Stay within the value-engineering target	\$175 or less in parts at quantity one, excluding cells (<code>project.yaml</code> , a hypothetical control target, not a limit). Restated in v0.4 from \$160 (CCK-DDR-001 item 4, decided by Amish, 2026-09-25)	Priced BOM
R13	Run from a safe supply	Certified 12 V DC adapter; nothing inside the unit above 13 V; no user mains wiring	Design review
R14	Limit unattended operation to safe stages	The self-discharge rest (no current flowing) can run without a person present; charge, resistance pulse and discharge run only with a person present. Redefined in v0.3 from "grading can run overnight without a person present" (CCK-DDR-001 item 2)	Design review; safety review by Amish
R15	Grade and match automatically	Grades per a configurable rule table; groups for any SxP pack with each parallel group's capacity within ± 1 % of the mean	Run on sample data

ID	Requirement	Target	Verification (TRL 3 or later)
R16	Keep a record for every cell	CSV row per cell with ID, source, measurements, grade, group and dates; export over USB or local Wi-Fi with no cloud account	Design review of the file format
R17	Be buildable and open	Off-the-shelf modules, no fine-pitch assembly; hardware CERN-OHL-S-2.0, software MIT	Design review of the parts list

Table 2. Status at TRL 3 (CCK-CAL-001 v0.3, paper estimates; at risk first, no requirement failed outright).

ID	Status	Basis
R5	At risk	12.0 cells per attended day if current stages finish the same day; 12.8 with an overnight pause; 8.0 for 2.5 Ah cells
R6	At risk	± 2.3 mV in the same channel (± 10.6 mV across channels, no longer allowed); cohort median cancels common relaxation, but the spread between cells is not bounded
R11	At risk	460 x 300 x 85 mm; 4.91 kg against 5 kg (constructable design, CCK-DDR-003)
R10	Not verifiable at TRL 3	Containment cannot be analysed credibly on paper; mean tray rise about 84 K for an assumed 100 kJ event
R1	Met (design review)	Eight holders for 18650 and 21700 cells
R2	Met	± 1.1 % without calibration, ± 0.4 % with
R3	Met (with per-channel calibration)	± 12.4 mV without calibration; ± 4.8 mV after it
R4	Met	0.83 m Ω per count; repeatability ± 2.09 m Ω worst case against ± 3.0 m Ω at 60 m Ω , with the pulse started only within 1 K of the bench (was ± 2.99 m Ω , at risk)
R7	Met (accuracy)	± 1.0 K at 45 °C, 1 Hz sampling; clip response time not verifiable at TRL 3
R8	Met	MOSFET cases 58.7 °C with fans; about 99 °C if the fans stop, so the firmware stops discharges on a fan fault
R9	Met (analysis)	Watchdog, undervoltage comparators, 3 A channel fuses and a series charge switch (CCK-CAL-001 Table 3); a detached thermistor is narrowed by a plausibility check but remains a residual risk
R12	Within the value-engineering target	\$172.40 estimated in parts, \$2.60 under \$175 (lines 19 to 21 added for construction, CCK-DDR-003)
R13	Met (design review)	Certified 12 V adapter; 12.6 V highest inside; 42.5 W of 60 W
R14	Met (design review)	As redefined in v0.3
R15	Met (synthetic data)	0.006 % group deviation for a 4S6P pack from synthetic cells
R16	Met (design review)	CSV per cell; lots map to the ReflowEconomy passport v0.2 with two schema gaps
R17	Met (design review)	Module-based build; open licences

■ Assumptions

- A typical salvaged 18650 holds about 1.8 Ah as found and arrives at about 30 % charge.
- A 14-day rest after a recharge that stops at 4.10 V under current separates cells with a harmful self-discharge from healthy ones; the 50 mV threshold is a starting value to be tuned with real cells.
- Operators remove cells from packs by hand before grading; disassembly is outside the scope of this design.

Safety: CellCheck charges and discharges salvaged lithium-ion cells, which can overheat, vent and burn. Charge and discharge run only with a person present. Meeting these requirements on paper does not make a unit safe, and grades do not certify any cell or pack.